

Director, User Experience, GE Aviation Digital

UX Portfolio

Anya Gerasimchuk

513.725.8035, avgera@hotmail.com

MY UX DISCOVERY PROCESS FOLLOWS

Design Thinking

WHO is the User?



- Persona Creation
- User Workflows/Journey Maps
- Empathy Maps
- Feature Prioritization Matrix
- Benchmarking
- Hierarchical Task Analysis
- Strategical and Actionable Research Findings

WHAT is Problem?



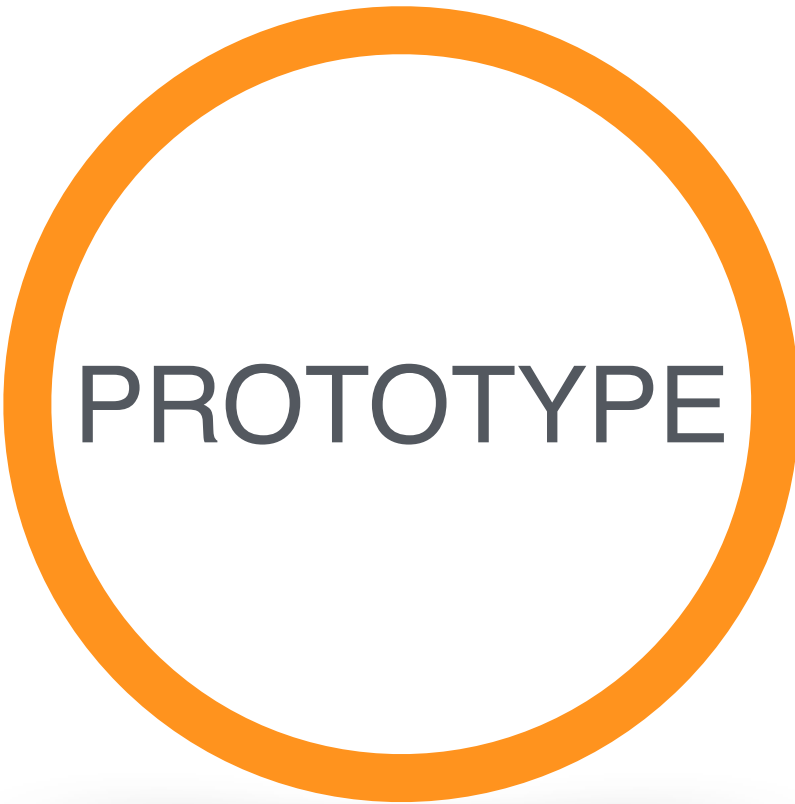
- Cognitive Walkthrough
- Task Analysis
- Card sorting
- Do Go Map
- Participatory Design
- Taxonomies and Metadata
- Mind Mapping

WHAT is the Experience?



- Site Maps
- Wireflows/Low Fidelity Wireframes
- Wireframes
- Storyboards
- Usage and Use Case Document
- Opportunities

HOW to enable the Experience?



- Heuristic Evaluation
- Usability Testing
- Experience Testing Walkthrough
- Perception & Value Proposition Testing
- Analytical Data Analysis



- UI Designs
- Recommendation Document
- UAT Testing Results
- Acceptance Criteria Results



EMPATHISE



DEFINE



PROTOTYPE



TEST

SKILLS & EXPERIENCE:

Persona Creation

User Workflows/Journey Maps

Empathy Maps

Feature Prioritization Matrix

Benchmarking

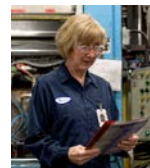
Hierarchical Task Analysis

Strategical and Actionable Research Findings

Persona Map

Beth Woods

The high-picture observer



Common Collaborators

- CPM, CTP

Data Needs (Inputs)

- Contract type, Product line and Customer
- Workscoped modules information
- Workscope history comparison

25 - 55 Y.O.

headquarters

5-15yrs
experience

Primary Tasks

- Finds workscopes of interest
- Reacts if the workscopes go beyond cost thresholds
- Preview and analyze the changes that forced workscope cost to go beyond threshold

Primary Needs

- Clear indicators of changes to workscopes
- Clear indicators of cost that goes beyond threshold
- Ability to preview reasons for cost increase

"I need to clearly see any changes that effect the cost threshold for all active worksopes I am responsible for. I need to understand which action has caused this"

Technology Touchpoints

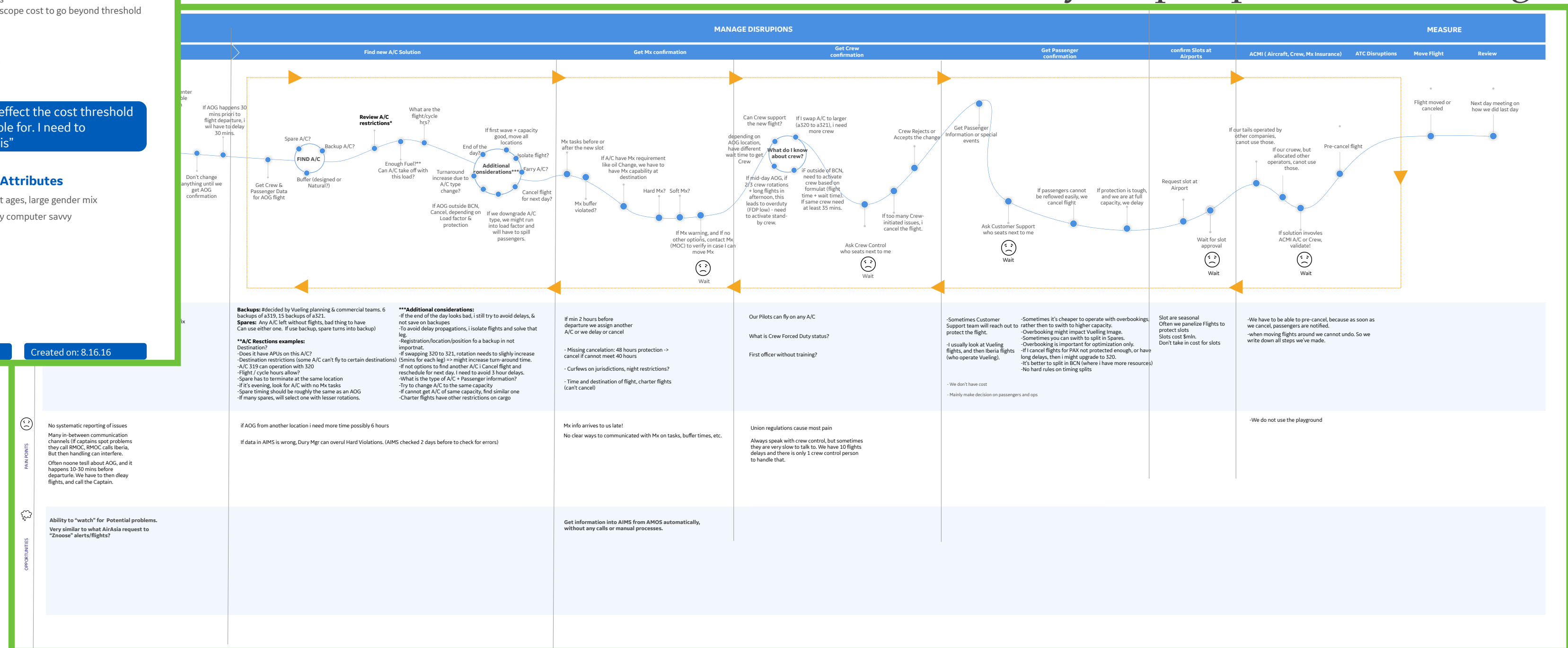
- Desktop, Tablet, Mobile Phone

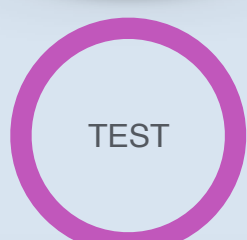
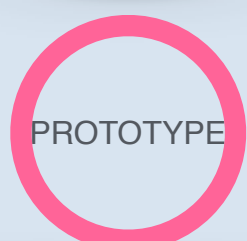
Other Attributes

- Different ages, large gender mix
- generally computer savvy

Created by: Anya Gerasimchuk

Created on: 8.16.16





SKILLS & EXPERIENCE:

Persona Creation

User Workflows/Journey Maps

Empathy Maps

Feature Prioritization Matrix

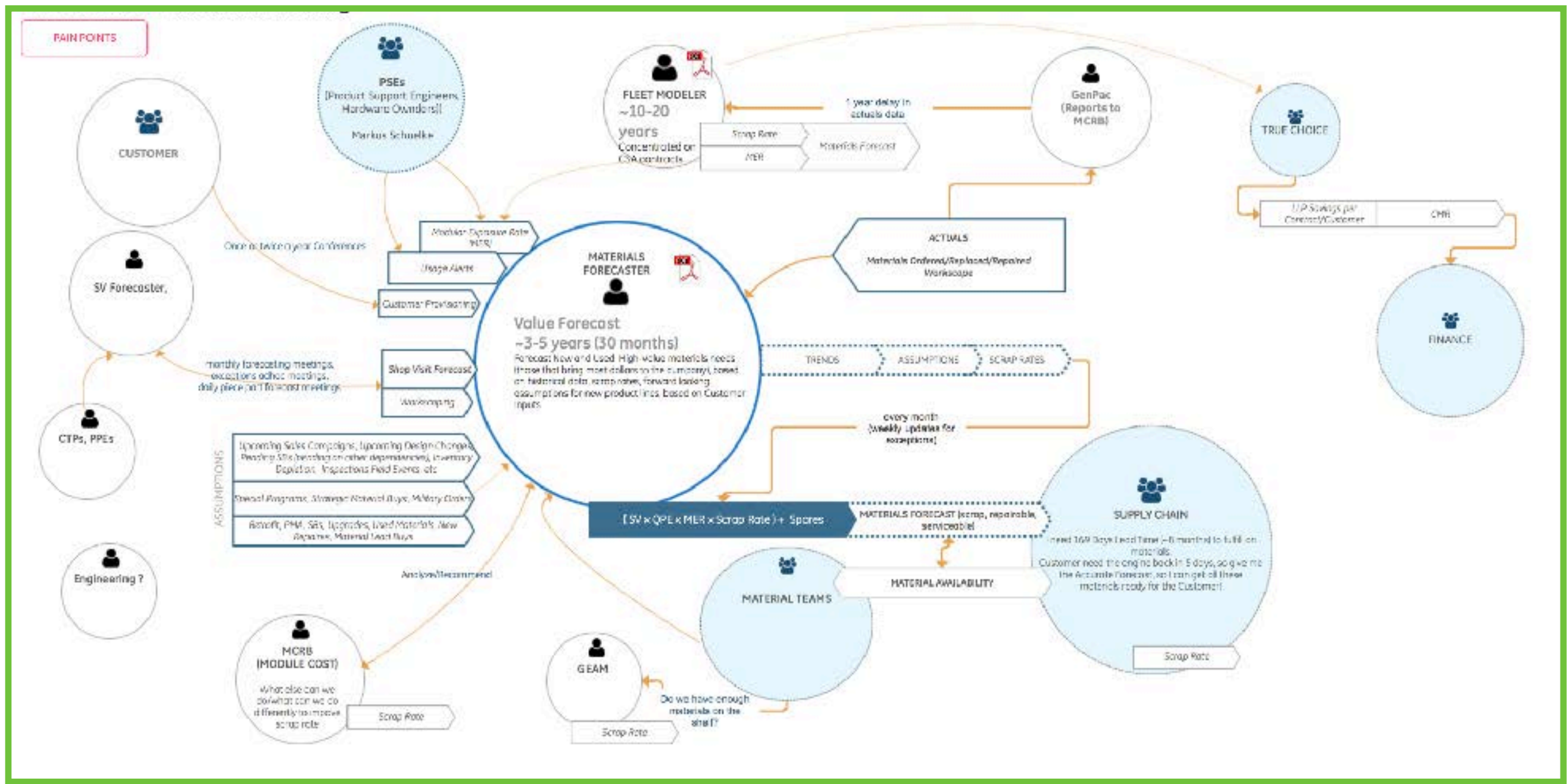
Benchmarking

Hierarchical Task Analysis

Strategical and Actionable Research Findings

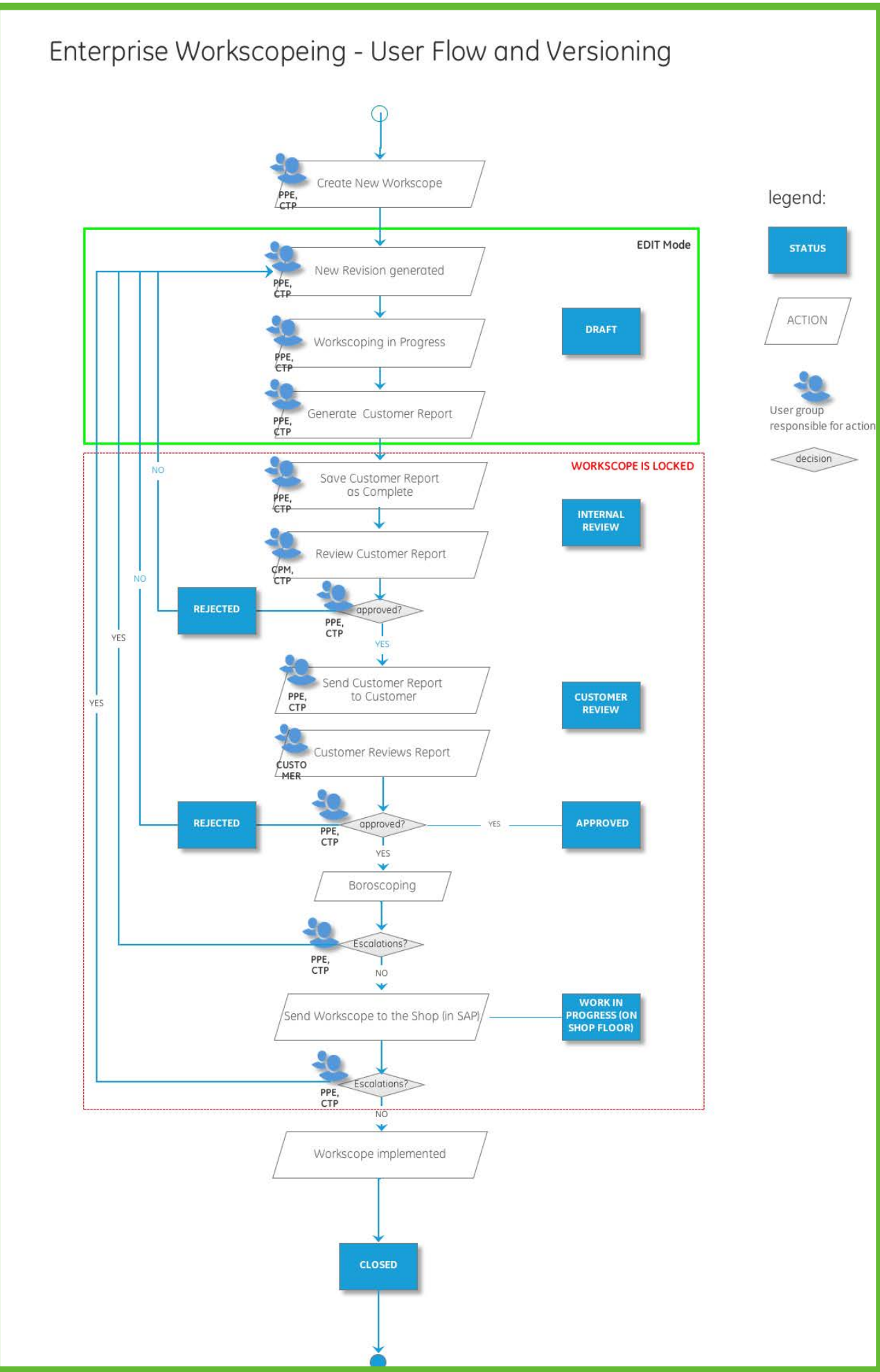
Ecosystem Maps / Task Analysis

This analysis explores how Material Forecasters at GE Aviation “predict” part needs for upcoming months, so that Supply Chain is ready for demand. It examines which players (teams) are involved, how our persona is dependent on them, what variables are needed for the algorithm calculations, etc. Via this analysis, we now came to a conclusion of which pain points need to be supported in our digital solutions to make sure that we simplify the process, versus adding yet another “app”.



Task Analysis

This example explored stages of the workflow to create a scope of work for a Jet Engine, and which personas participate in each step. The example below concentrated on the customer requesting work to be done, to the finalization and agreement on the final scope of work, to be provided to Shop Floor mechanics.



EMPATHISE

DEFINE

IDEATE

PROTOTYPE

TEST

SKILLS & EXPERIENCE:
Cognitive Walkthrough
Task Analysis
Card sorting
Do Go Map
Participatory Design
Taxonomies and Metadata
Mind Mapping

Participatory Design

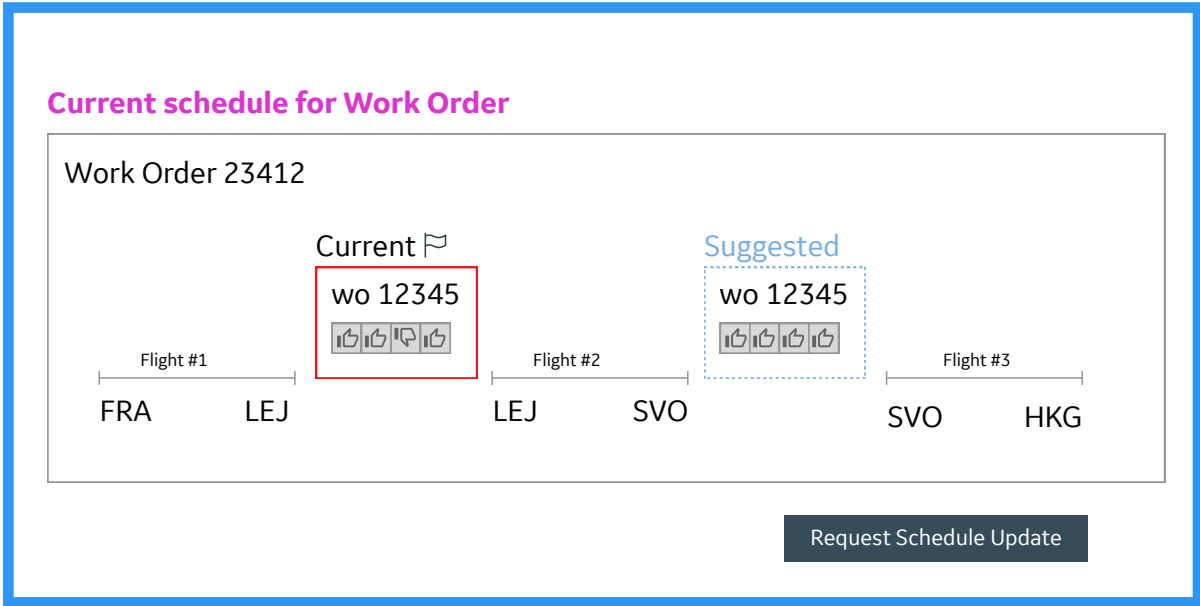
One of my favorites, **participatory design**, allows designing with customers to get a deeper understanding of the problem space & to get trust. This is the fastest and the best way to define the problem by allowing users lead us through their thinking process. Here, breaking into 3 groups, we sketched the “ideal” flow of how maintenance schedulers at a Cargo Airline should streamline maintenance scheduling procedures to eliminate flight disruptions.

All Tails

Search

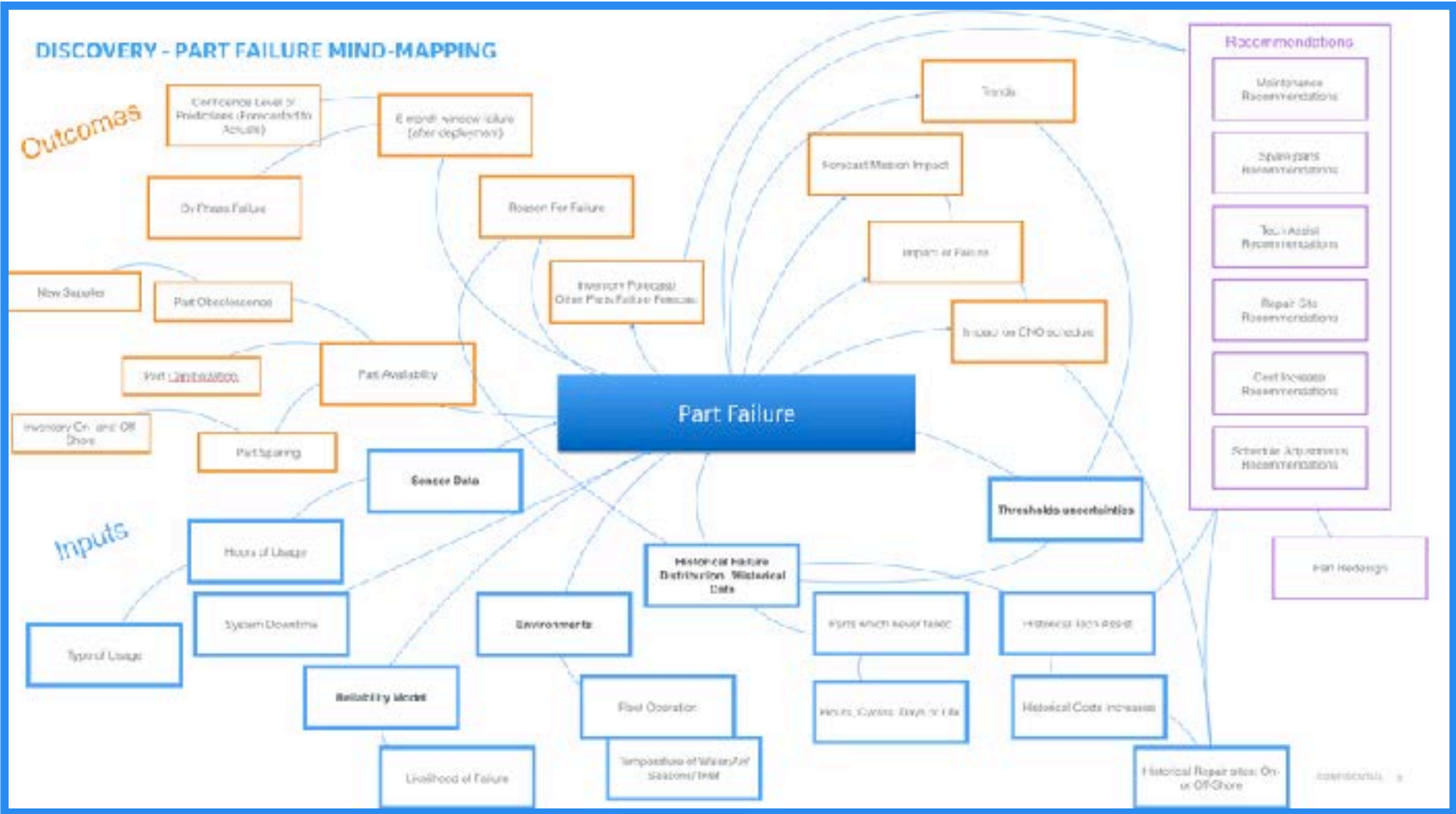
Filter

TAIL NO	DETAILS	LOCATION	STATUS
	Boeing 737	SVO-LPG	Due Date
	Boeing 737	SVO-LPG	Parts
	Boeing 737	SVO-LPG	
	Boeing 737	SVO-LPG	
	Boeing 737	SVO-LPG	
	Boeing 737	SVO-LPG	
	Boeing 737	SVO-LPG	
	Boeing 737	SVO-LPG	
	Boeing 737	SVO-LPG	
	Boeing 737	SVO-LPG	



Mind Mapping

More complex problems might need more detailed definition. The below **mind-mapping** exercise was done to determine the possible inputs into the “part failure” prediction algorithms. This analysis was done with Data Science in mind. To establish user trust, Data Science problem space needs to be understood and results linked with easy visualizations based on real use-cases.



EMPATHISE

DEFINE

IDEATE

PROTOTYPE

TEST

SKILLS & EXPERIENCE:

Site Maps
Wireflows/Low Fidelity Wireframes
Wireframes
Storyboards
Usage and Use Case Document
Opportunities

Storyboards & Use Cases

Sample Use Cases

All cases are examples only. We drive Use Case creation based on the unique Customer Needs.

1. Smart scheduling into Shops based on historical data and analytics



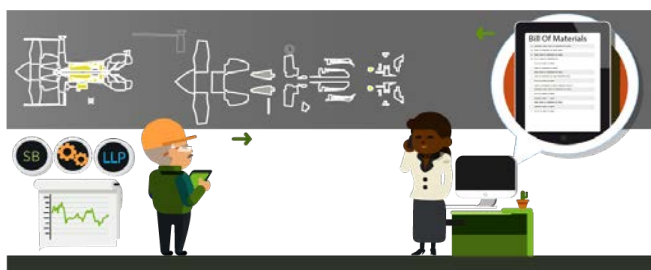
Capacity Planner

Determine urgency for upcoming inductions

I need to induct into my shop 7 engines. For that, I need to know who is the Customer, what needs to be done to the engines, if there are any exceptions.

I need a confirmation on the work that needs to be done with the Scope Engineer.

2. Improved Workscope productivity for Scope Creep & Cost tracking, and Materials management



Scope Engineer

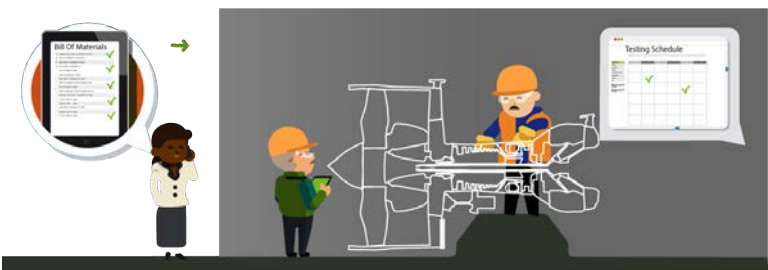
Review high-level scope of work

What are the Serial Numbers? I then can look into health, Configurations, any Specs, and perhaps there are recommended actions based on historical observations.

OK, based on all this knowledge I have a better picture of what is upcoming, and what type of work we might expect.

Capacity Planner, 5 of these engines will need Heavy Scope done on them, and 3 out of those 5 need to be turned around very quickly. The other 5 are Medium Scope.

3. Intelligent parts and WIP tracking to speed up Assembly and Testing



Materials Mgmt

Review Bill Of Materials, based on Workscope, and order

The workscope is ready, inspections are done. I know which Materials or Parts needs Repair, and which ones I need to order, whether Used or New. Let me look at all available options that fit my selection criteria.

Materials Tracking

Tracking material processes

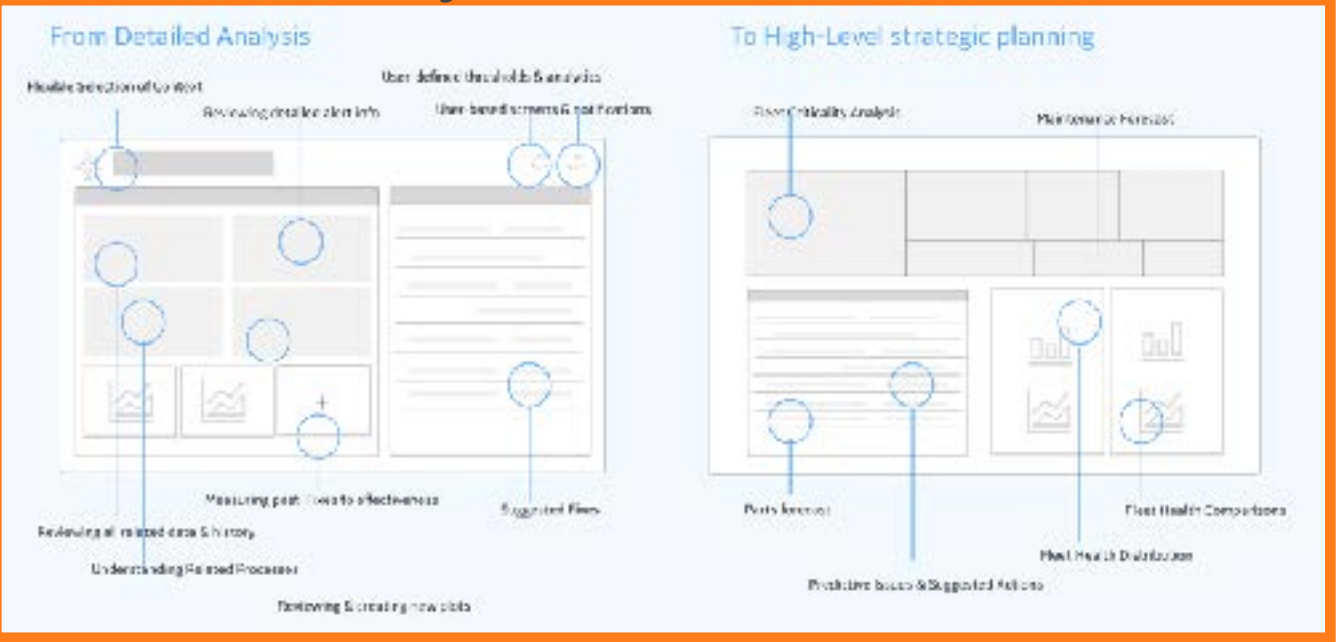
Now that I have BOM, I need to make a decision on whether to repair or order new or used materials, and then observe how materials flow into the shop. I need to be able to analyze and have intelligence into all repair processes work, or if I need to reorder anything.

Assembly and Test

Materials onsite - Assembly and Test

With the streamlined flow of materials, I can start Assembly and then Testing. To get the asset out of the Shop, the coordination of testing is very important.

Low Fidelity Wireframes



Enterprise Workscoping

Digital MRO

Dashboard

WORKSCOPE 234566 ESN #876455

VERSION 1 - INITIAL LPT S6 Blade Interlock wear Alaska Airlines, GE90-M10 CSO: 0, TSN: 40369, TSO: 0, CSN: 18095

TARGET COST 1.4min PROJECTED COST 1.5min ACTUAL COST

72-00-00 ENGINE ASSEMBLY (M10)

MODULE 72-00-00 Engine Assembly (M10)

UPDATES LEVEL LLP SB Remove

72-21-00 Rotating Hardware (M10) 1 2 3 4

72-00-06 Fan Case Assembly (M15) 1 2 3 4

72-00-25 Lower Bifurcation (M80) 1 2 3 4

72-00-00 Base Engine Assembly (M20) 1 2 3 4

72-00-01 Fan Booster (M40) 1 2 3 4

72-21-00 Fan Rotor Stator Assy (M21) 1 2 3 4

72-22-00 No1R Bearing Assy (M22) 1 2 3 4

72-26-00 No2B Bearing Assy (M26) 1 2 3 4

72-23-00 Fan Frame Assy (M31) 1 2 3 4

72-00-00 ENGINE ASSEMBLY (M10)

SERIAL# TSN, CSN, JSO, CSO

LEVEL: Level 1 2 3 4

Level Instructions Split into Fan and Propeller modules

Special Instructions/Notes

EMPATHISE

DEFINE

IDEATE

PROTOTYPE

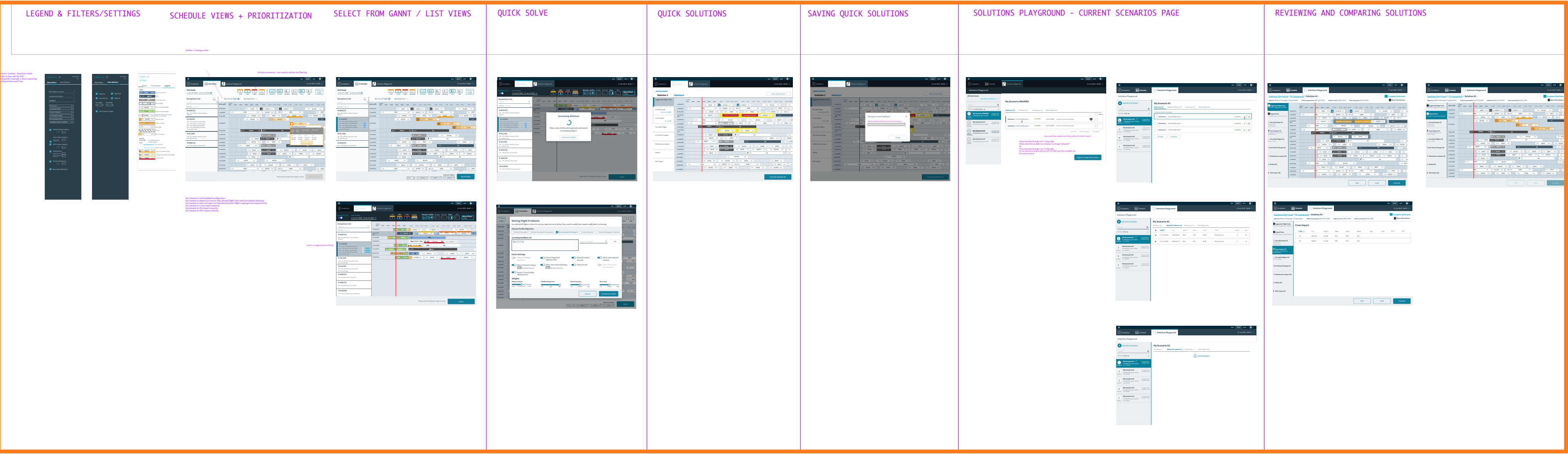
TEST

SKILLS & EXPERIENCE:

- Site Maps
- Wireflows/Low Fidelity Wireframes
- Wireframes
- Storyboards
- Usage and Use Case Document
- Opportunities

Conceptual Wireframes

Once space is understood and empathy to end-users is built, it’s time to ideate on possible solutions. The below is an example of a flow to allow Duty Manager automate the process of flight disruption management. This can be done in a lower-fidelity storyboard format, or as depicted below, in a higher-fidelity style where details are incorporated to explain the complexity of the decision making process.



EMPATHISE

DEFINE

IDEATE

PROTOTYPE

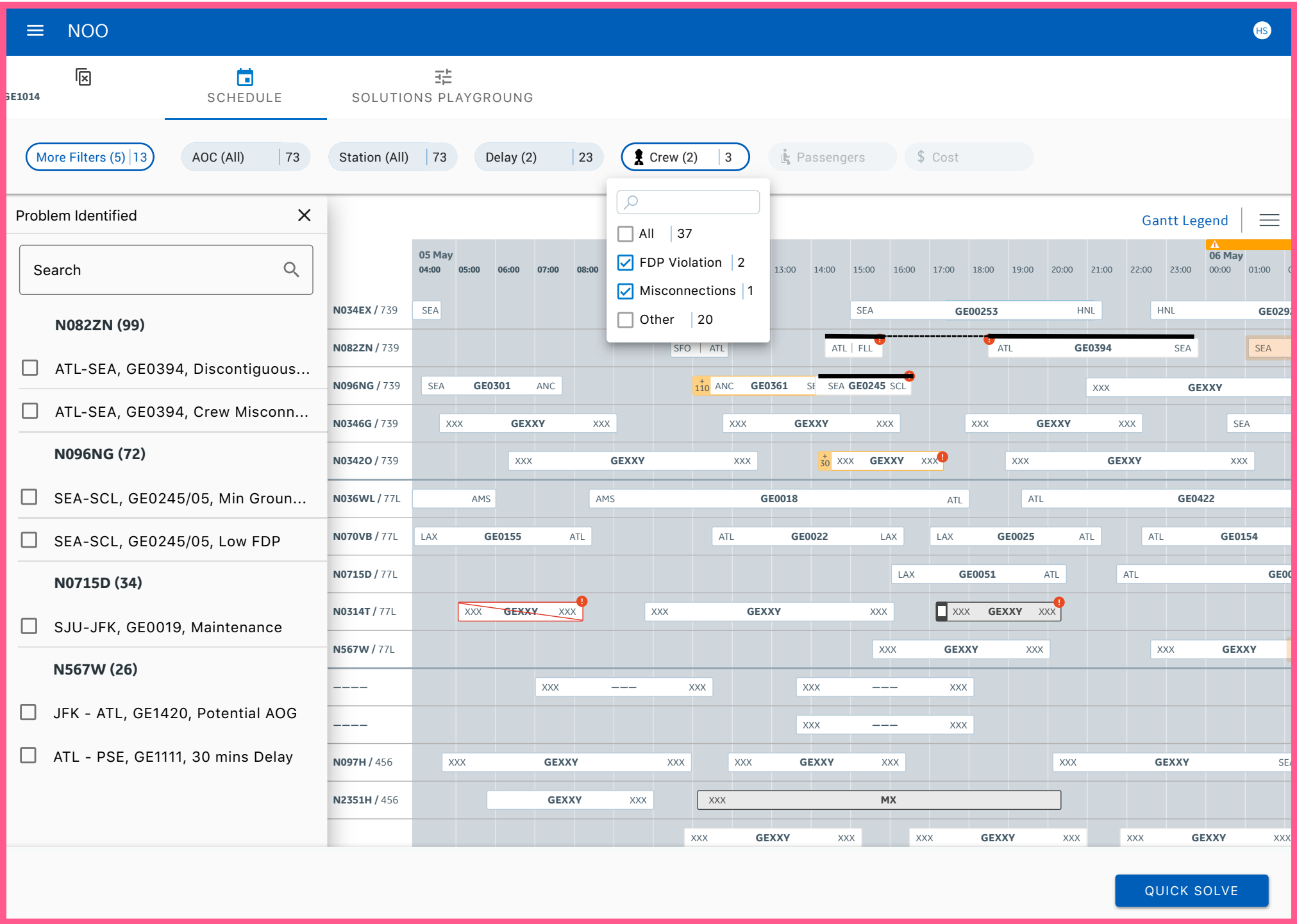
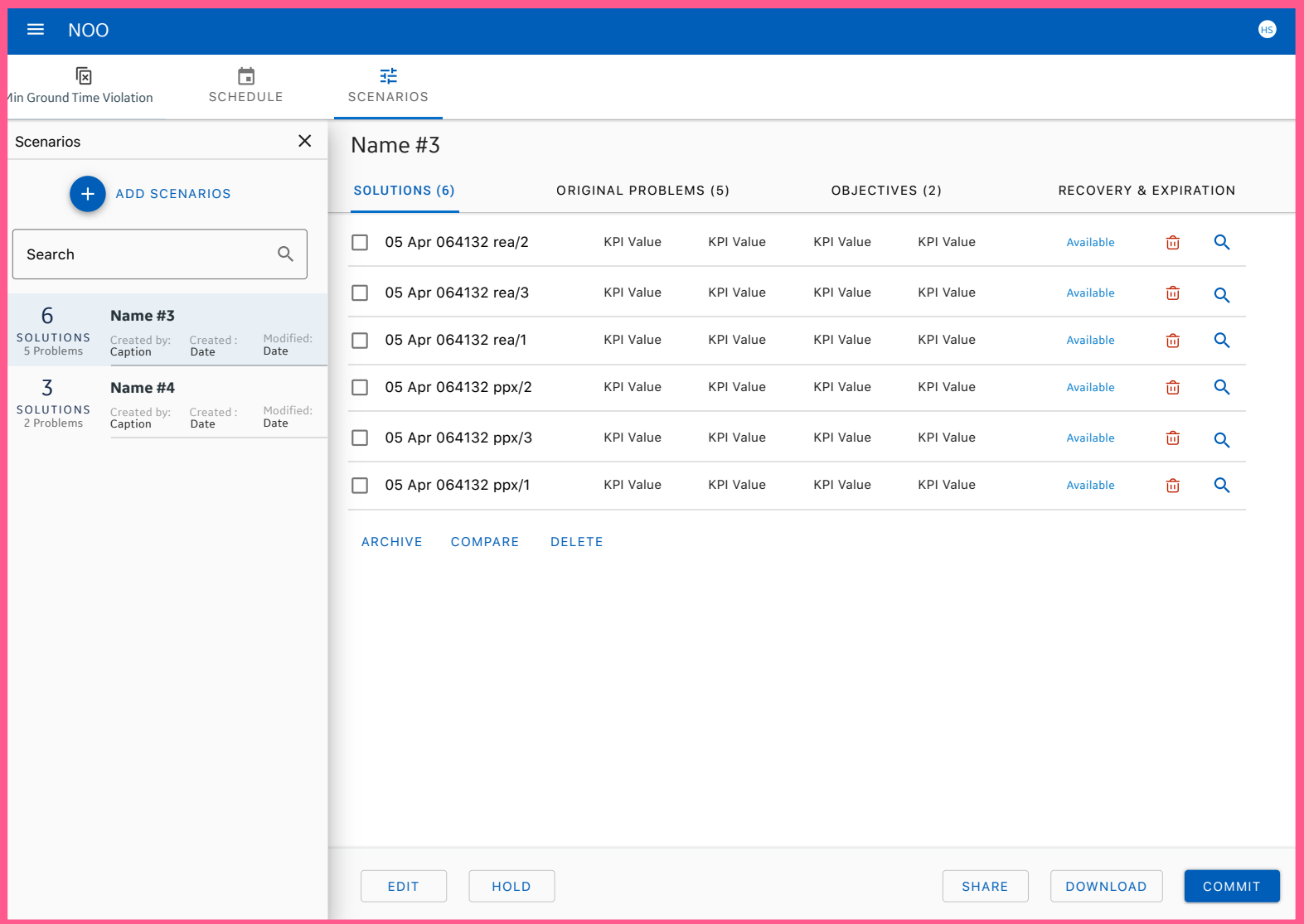
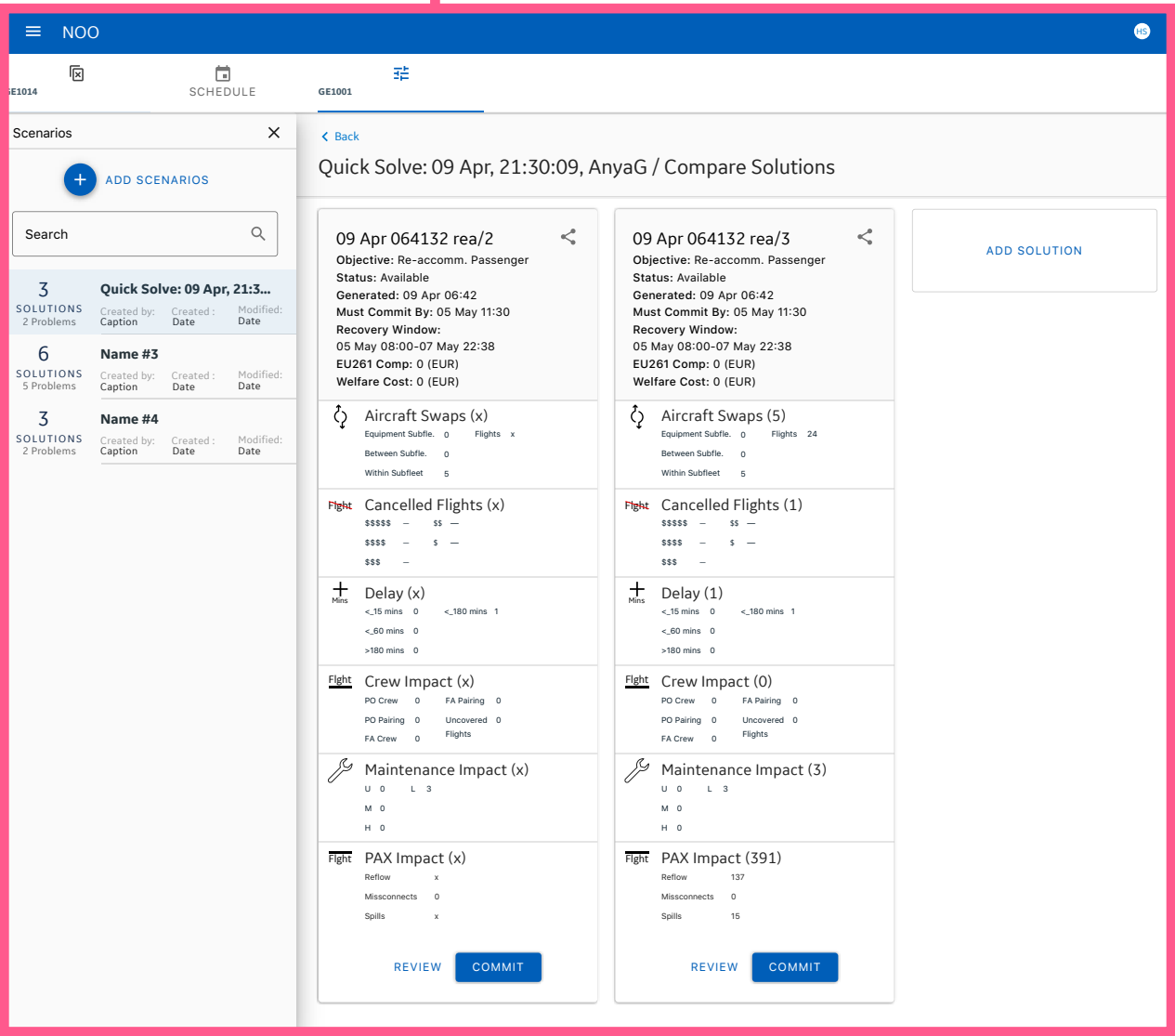
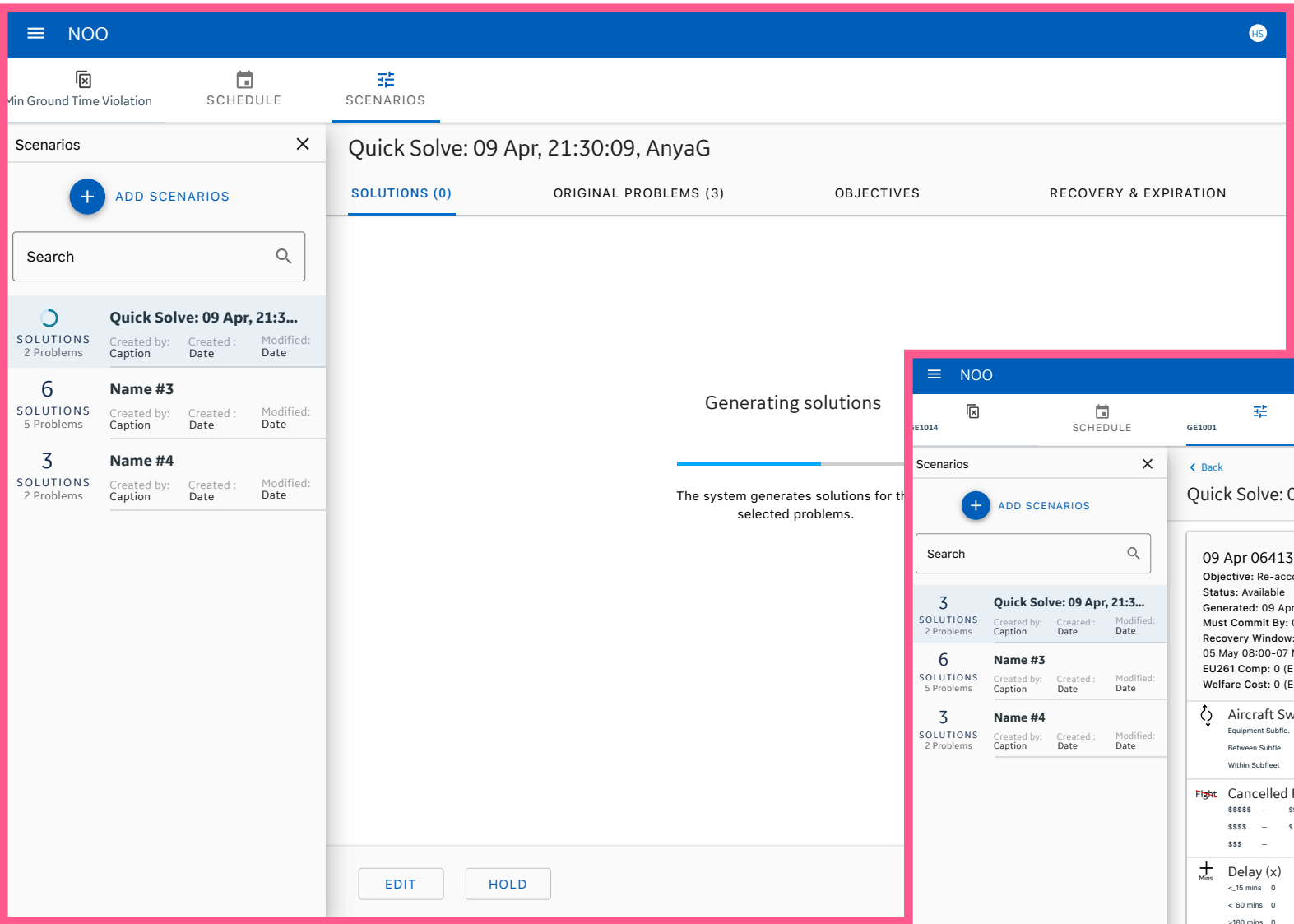
TEST

SKILLS & EXPERIENCE

- Heuristic Evaluation
- Usability Testing
- Experience Testing Walkthrough
- Perception & Value Proposition Testing
- Analytical Data Analysis

Flight Disruption management.

The application displays flight data real-time, which combines the Schedule for networks operations, and the possible disruptions for a particular time period. The user able to review disruptions in either Gantt or List View formats, easily prioritize work, and start solving. The application is based on the “solver” algorithms that take various variables into calculations of the best solutions, which are then presented to the user. We identified that such automation of work presented adoption challengers to our user communities, where the solutions were viewed as coming out of the “black box”, therefore we used careful analysis of how user experience can aid in trusting the solutions and thus allowing for better adoption.



EMPATHISE

DEFINE

IDEATE

PROTOTYPE

TEST

SKILLS & EXPERIENCE

Heuristic Evaluation

Usability Testing

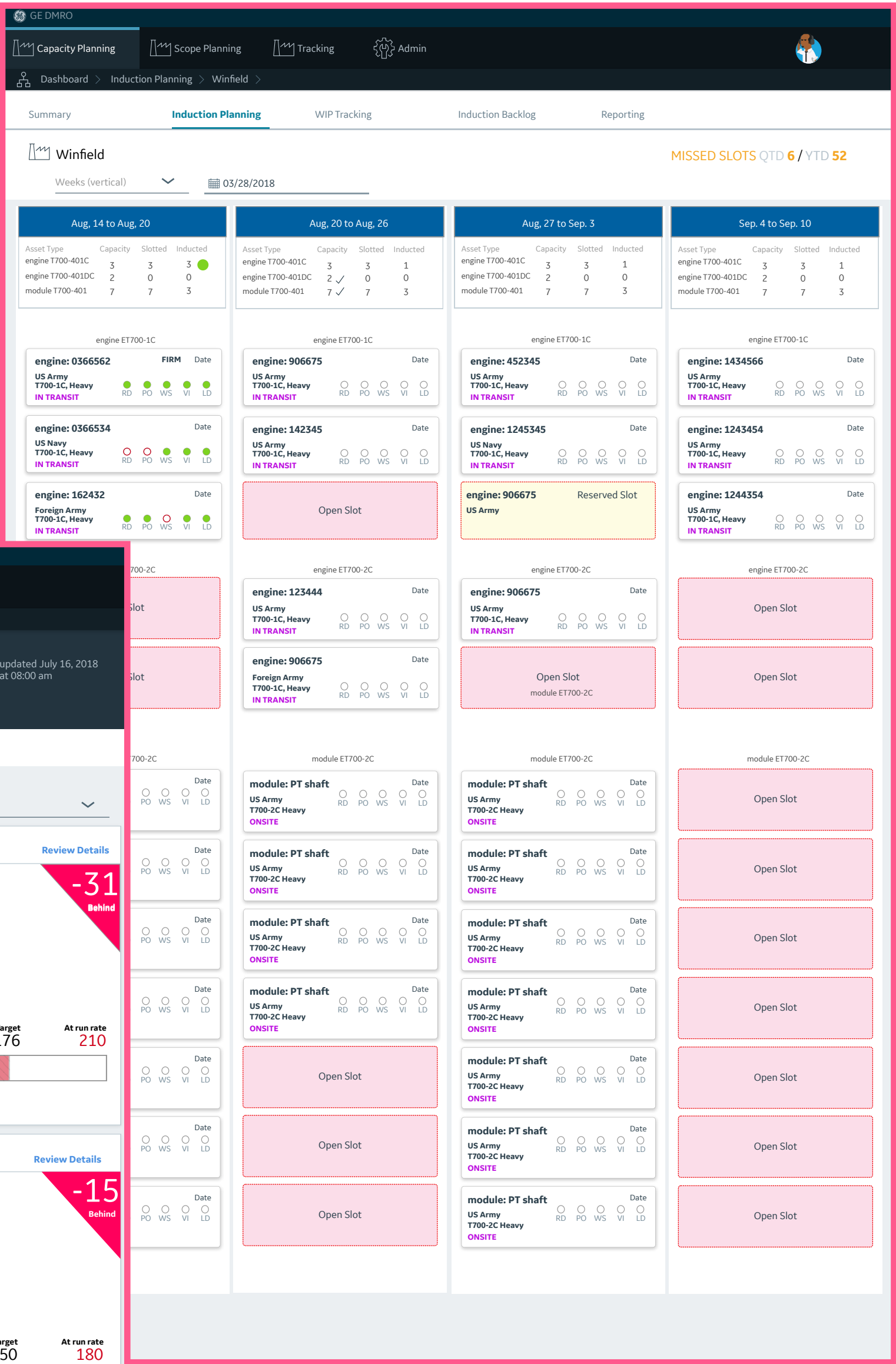
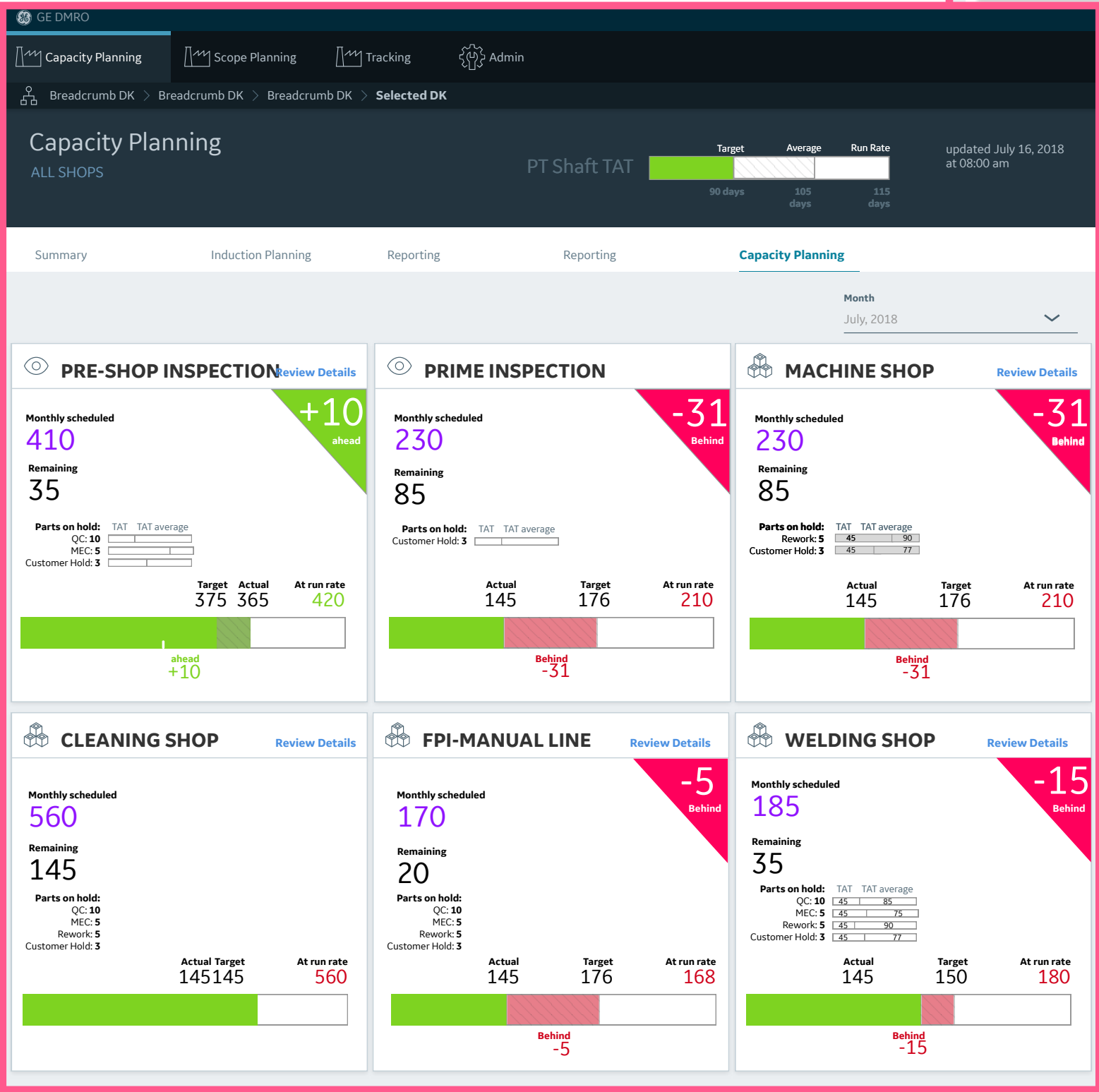
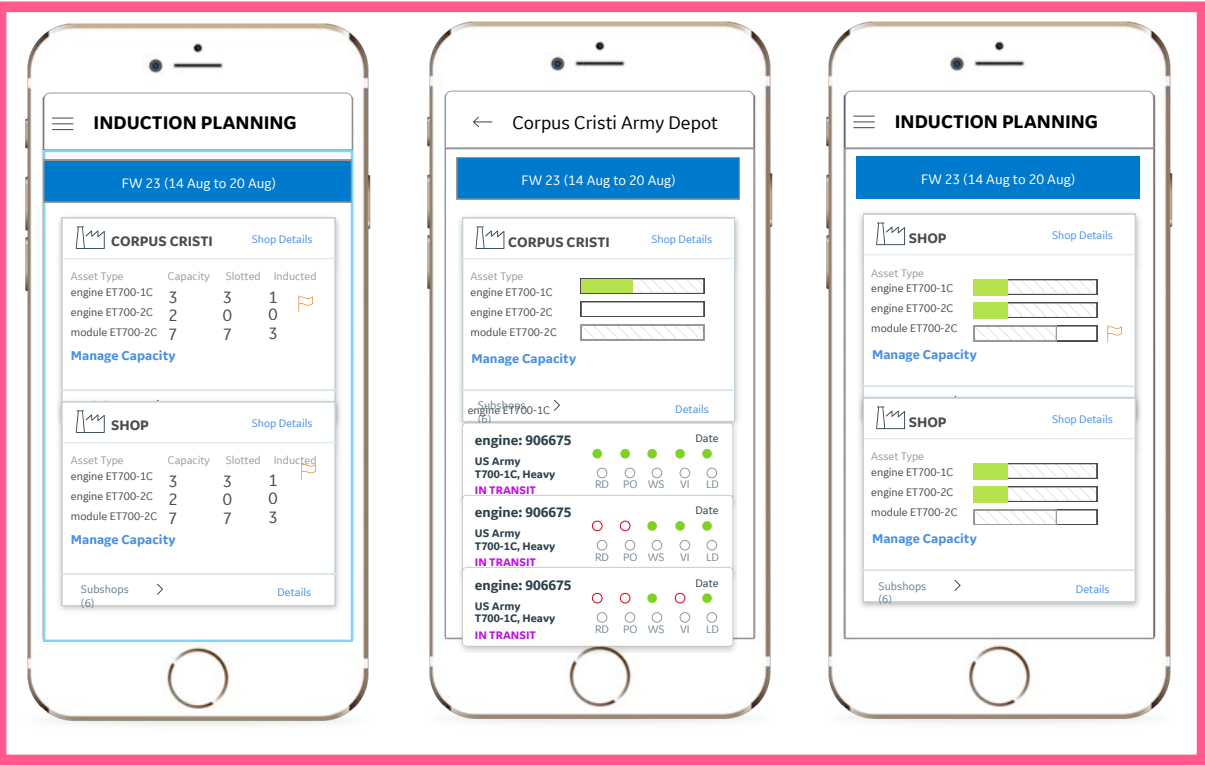
Experience Testing Walkthrough

Perception & Value Proposition Testing

Analytical Data Analysis

Capacity Planning.

Maintenance Repair and Overhaul shops (MROs) throughout the world face a challenge of capacity planning, in order to service (or build) million-dollar equipment, like jet engines, aircraft bodies, and other “hard-core” items. MROs have complicated processes to forecast, prioritize, plan and carry out work to satisfy customers. The capacity planning application saved millions of dollars to internal GE MRO shop floors and external customers, by providing insight into the upcoming “slot” schedule, needed skill sets and materials, required compliance check points. The application provided capability of scheduling work and also reviewing capacity statistics on a high level to see progress and areas of improvement.



EMPATHISE

DEFINE

IDEATE

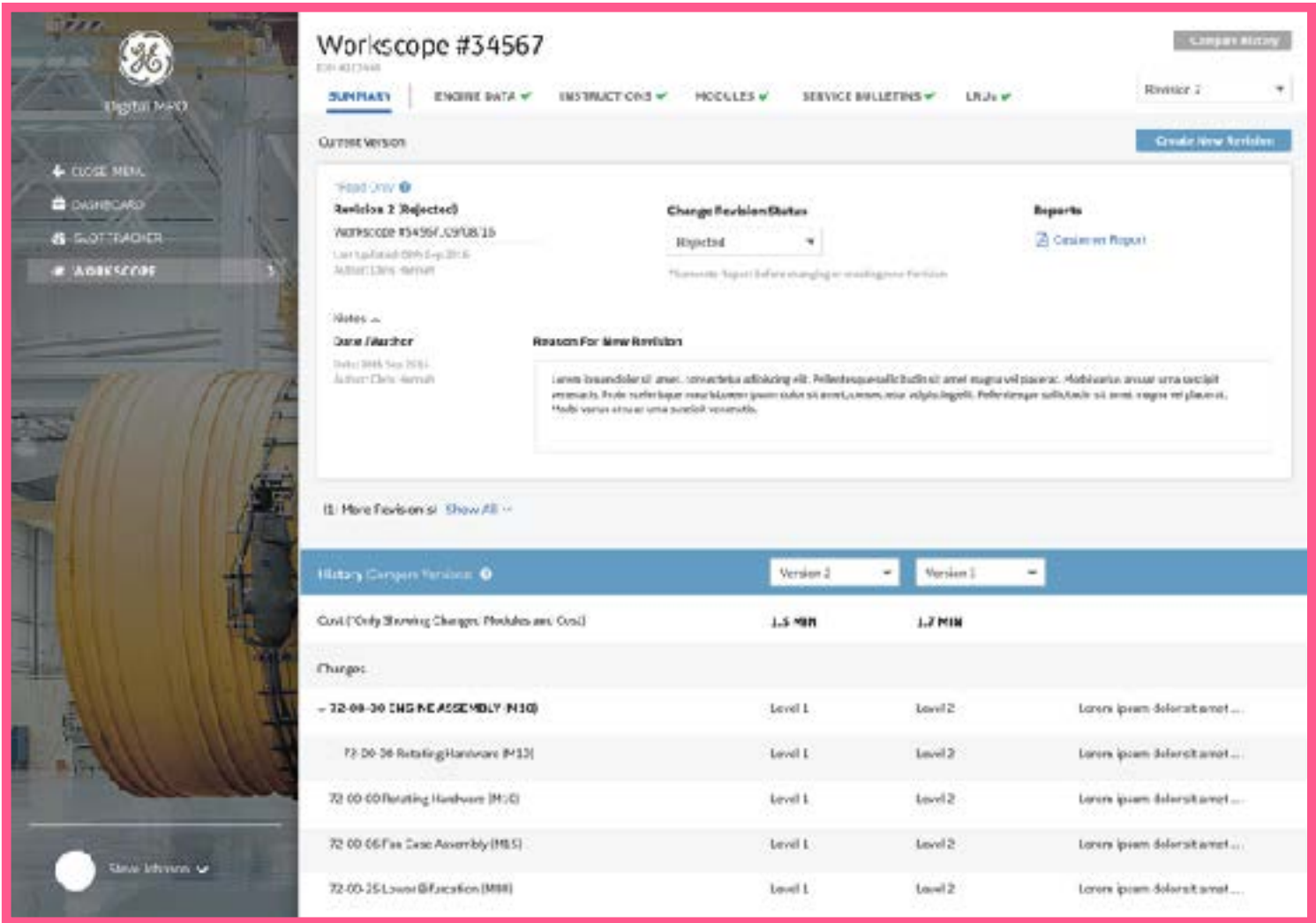
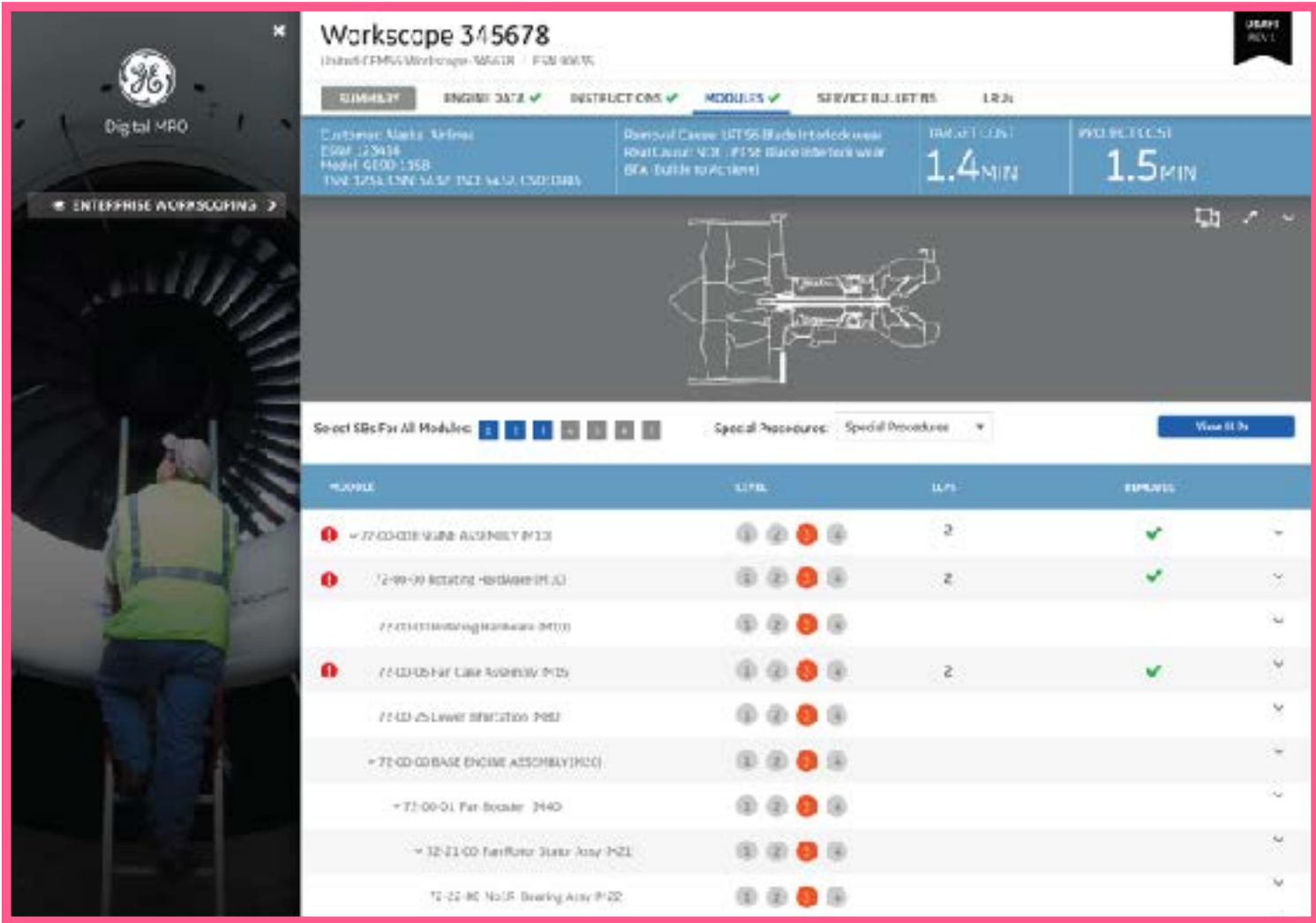
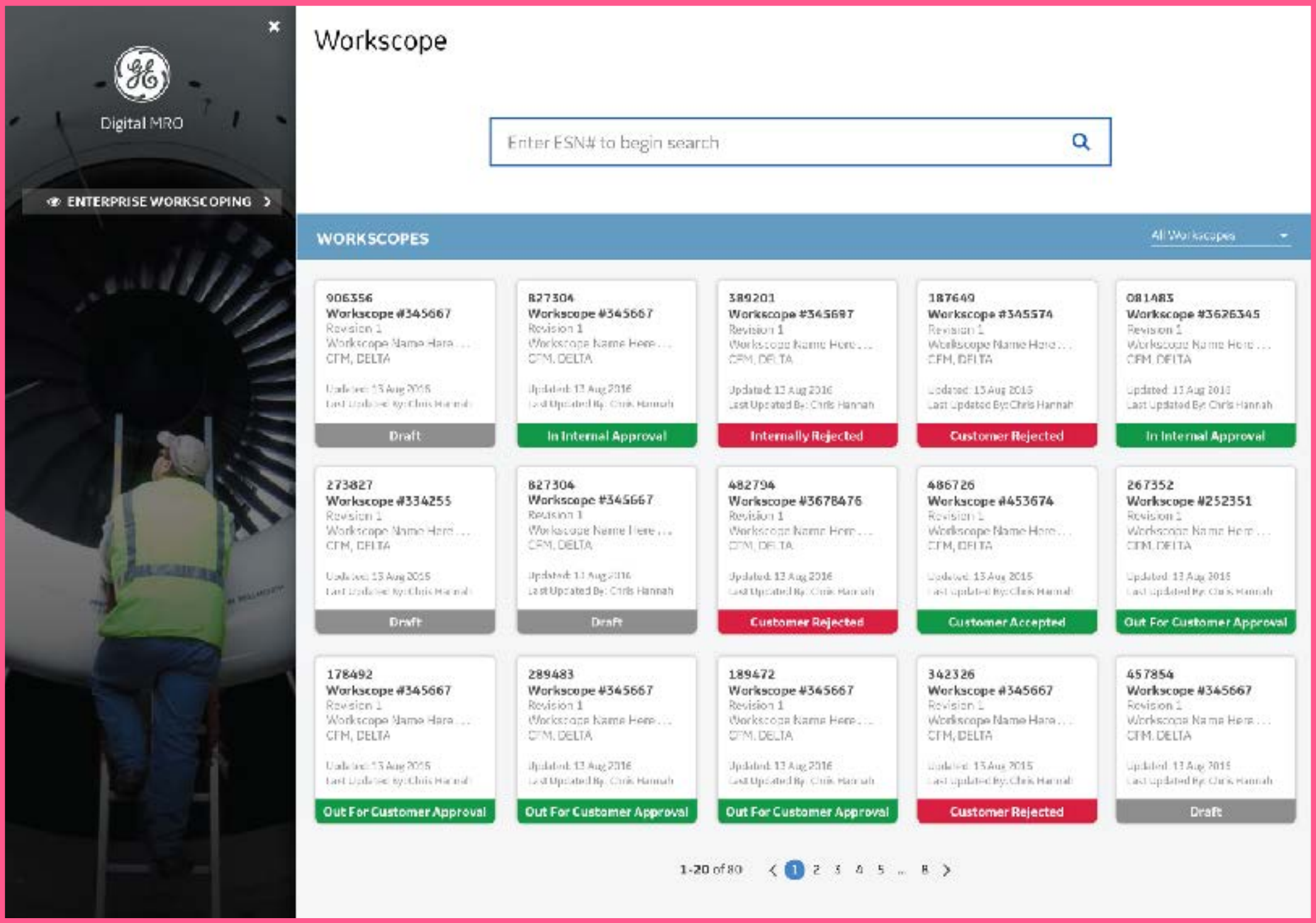
PROTOTYPE

TEST

SKILLS & EXPERIENCE

Heuristic Evaluation
Usability Testing
Experience Testing Walkthrough
Perception & Value Proposition Testing
Analytical Data Analysis

Enterprise Workscoping application is in the heart of GE Aviation services business. The application became a must go to for engineering “customer technical program” communities responsible for setting the scope of work for servicing engines and other hardware, and communicating the decision to the customer. The application aggregates many data sources to allow view engine’s past history, maintenance records, provide health analytics, review service bulletins, and provides necessary mechanisms for versioning, approval workflows and communication.



EMPATHISE

DEFINE

IDEATE

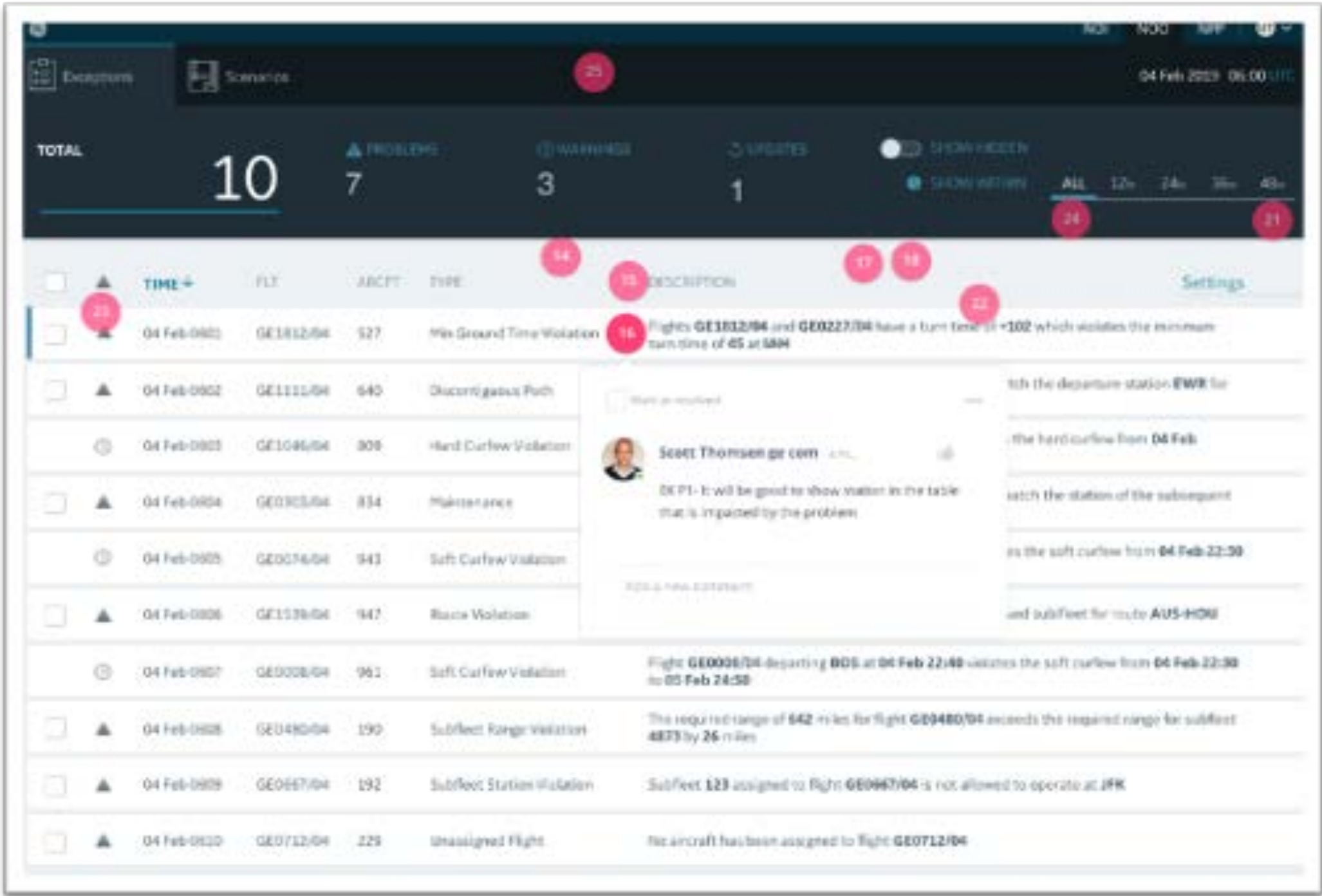
PROTOTYPE

TEST

SKILLS & EXPERIENCE

UI Designs
Usability testing
Recommendation Document
UAT Testing Results
Acceptance Criteria Results

Usability Study example allowed us to observe how a user would interact with an application. Same study was done in a moderated and a non-moderated way. The example on the right is a written result of both types of studies (we followed up with a re-designed screens, shown in the “prototyped” examples in the previous pages of this document). The results of the study where then broken into user stories, to be prioritized into agile sprints for development. I like to repeat the test consistently, and in the ideal scenario - every 6 weeks as the development proceeds, if we have access to real users.



For security, the screenshot is pixelated.

Exceptions Protocol

Tasks

1. Review the screen.
2. Read the summary drawer
3. Read the table
4. Select 3 problems, GE1812/04, GE1111/04, and GE1300/04
5. Solve the selected problems.

Hypothesis

The user gets the highest view of Exceptions and is able to select a problem to solve. However, they might be confused with Problems, Warnings, and Updates
-What are you seeing at the very top of the screen and what does it represent?
-What do you expect to see when you select the table?
-What are the numbers telling you?
-What are you seeing in the table below?
-What actions do you think you would perform from this screen?
-Probe for expected functionality of search, filters, hide, add, solve, expansion of table row
-What types of problems do you normally solve for?

Top questions to be answered

1. How do you prioritise problems?
2. What information is required to make a decision?
3. How many problems do you typically have?
4. What types of problems do you have?

Exceptions Results

Emirates (EK) 11/2018

- Both the participants were able to perform the tasks easily
- At first, participants didn't grasp the idea of problems and warnings, they were not clear what to do with warnings
- Participants needed help to understand what does update number mean
- Further dividing problems into different levels of priorities may help identify which problem to solve first, however, business rules will be needed to create priorities for different customers
- Show additional information related to the problem, may be by expanding the row

Royal Air Maroc (RAM) 02/2019

- Participants tried to select a problem by clicking anywhere inside the row
- PI asked to show app live status

Pet Observations 02/2019

- Can we show traffic lights to indicate the app is working. On hotline this is often a problem and sometimes need to force a refresh
- NOD exceptions screen has no impact assessment to help the user prioritise exceptions

Evaluating NOD solutions is very hard. Our example suffered as there is no financial info in the NOD results. For each business objective it would be nice to have a comparator of how well a solution meets the objective (benefits) and then secondary info on summary impact on Pass, Crew, Maintenance (cost or complexity of implementing the solution) then further info on solution detail. Easy to say, perhaps harder to do

Verling TBO

Issues

- Prototype was not used for the RAM testing.
- Coherent story/tasks end to end across different screens
- We can provide participants some initial training and then ask to solve for certain tasks
- Too long ... it took about 90 min for each study and generally after 1hr into the study participants start to lose interest. However, this is tricky as it's not straight forward to get multiple slots on their busy schedule
- Moving forward, it will be useful for advance we identify and agree on study objectives with the product team. This will help set the expectations related to the outcomes.
- Based on the objectives we need to identify proper protocol. For example, the current qualitative analysis works to collect participants high level feedback regarding the tool capabilities. However, we also need to think about comparative/quantitative analysis to generate insights on different design options.

Wagas 3/08/2019

Notes

Not testing /hide/show

I think it would be useful to include some context in the report, where it may influence the results. For example the IT systems they are currently using to obtain info, the disruption problem complexity (flights per day and a brief description of the RAM network); the pain points currently - Per 2/2/2019

Example explanation to our user-participant in the study:

This test will be done as a Moderated Usability test. You will be a “user” (a duty or an ops mgr) pretending this is the “day of ops” and you need to manage disruptions using our application. (Please note that the application is still in the design stage and is in a form of flat design screens with minimum interactions and fake data). A User Experience designer will be a “moderator” and will be asking you questions targeting to test if the solution we provide is usable & functional and listening to your responses. The GE team (PMs and UX) will also be listening, taking notes, and providing questions to the moderator.